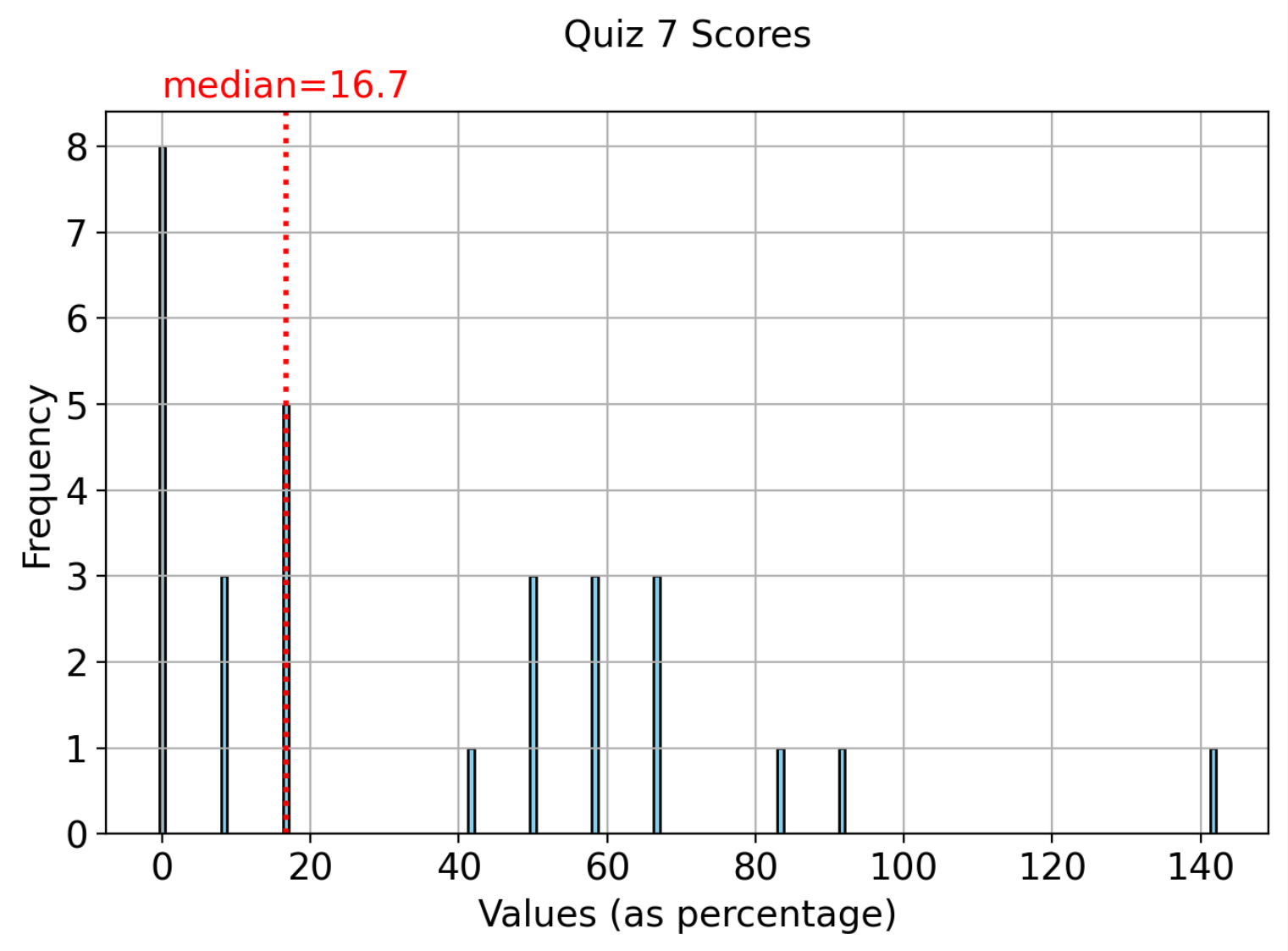
10/28/2026: Big O, Big Omega, and Big Theta Notations

CSCI 246: Discrete Structures

Textbook reference: Ch 11.2, Epp

Today's Agenda

- Mini-lecture ($\approx$ 30 mins)
- Group exercises ($\approx$ 20 mins)


Quiz 7 Scores
median=16.7
Frequency
Values (as percentage)
0
1
2
3
4
5
6
7
8
0
20
40
60
80
100
120
140

Study Guide For Quiz 8

Group exercises. Know how to solve the group exercises from the course meetings on Induction (10/14) and Recurrence (10/16).

Readings.

Sec. 22 (Induction)

- Know how to use Theorem 22.2 / Proof Template 17 to prove Props. 22.3 and 22.5.
- Understand the difference between Proof Template 17 and 18.

Sec. 23 (Recurrence)

- Know how to solve first-order recurrences from Props 23.1 and 23.3. For example, see Example 23.2.
- Know how to solve second-order recurrences from Theorems 23.5 and 23.9. For example, see Example 23.6.
- Know how to use Theorem 23.17 to find a polynomial expression to generate the first few terms of a sequence, as in Example 23.18.

Important Note. The max of your quiz 7 and quiz 8 scores will be used for both.

Mini-lecture:
Big O, Big Omega, Big Theta Notations

Which would you rather have?

Option A

$1,000,000 on Nov 1

Option B

$1 on Nov 1,
doubled by the
end of every day

Which would you rather have?

Option A

$1,000,000 on Nov 1



Hopper chooses Option A.

Option B

$1 on Nov 1,
doubled by the
end of every day



Bob chooses Option B.

Which would you rather have?

Option A

$1,000,000 on Nov 1



Hopper chooses Option A.

On Nov 30:

$1,000,000

Option B

$1 on Nov 1,
doubled by the
end of every day



Bob chooses Option B.

On Nov 30:

$1,073,741,824

Why?

Why?

Let n be the date in November, $f(n)$ be the value of your bank account on date n.

Why?

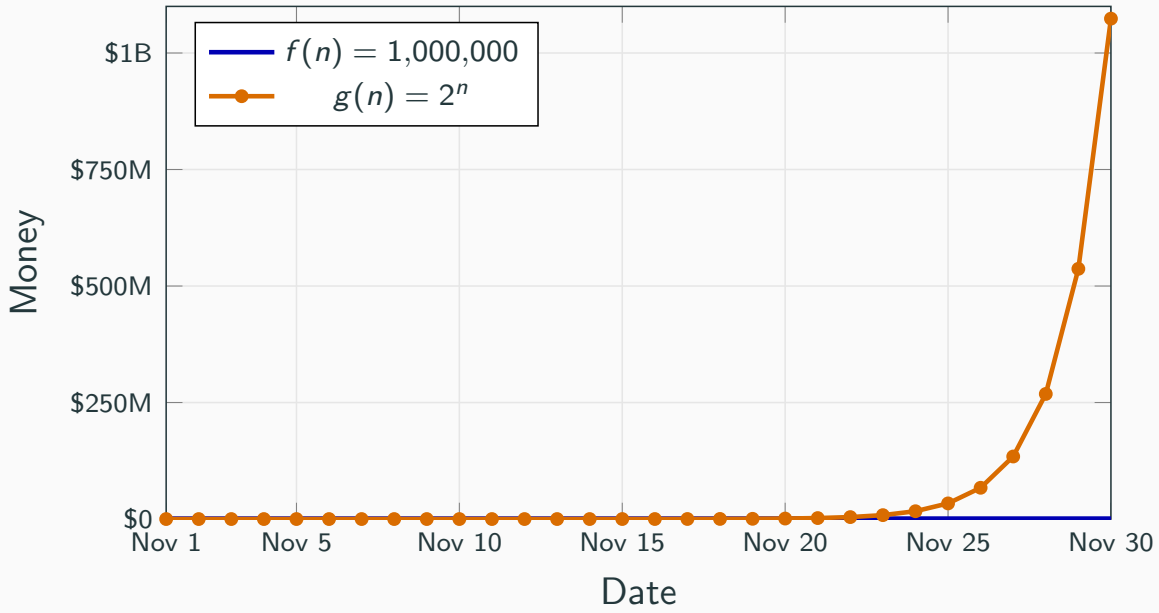
Let n be the date in November, $f(n)$ be the value of your bank account on date n.

Option A

$$f(n) = 1,000,000$$

Option B

$$g(n) = 2^n$$



$g(n) = 2^n$ is eventually greater than $f(n) = 1,000,000$.

Why?

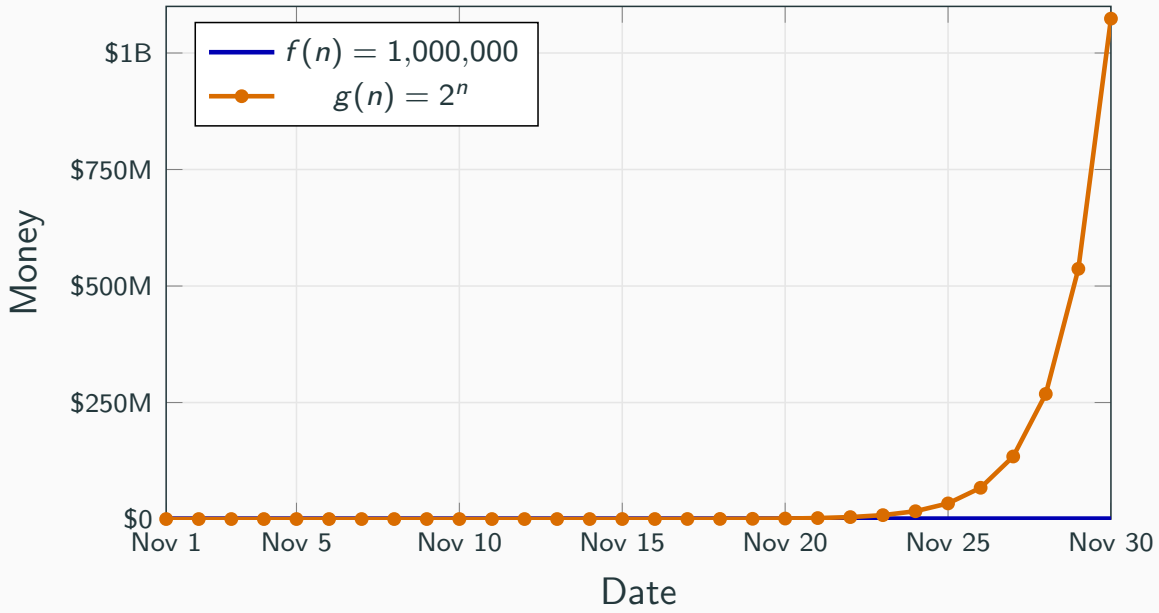
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Option A

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Option B

$$g(n) = 2^n$$



$\underbrace{g(n) = C2^n}_{\text{exponential}}$ is eventually greater than $\underbrace{f(n) = D}_{\text{constant}}$.

Which algorithm would you rather apply to a large dataset?

Option A

Passes through each sample 10,000 times.

Option B

Passes through each pair of samples once.

Which algorithm would you rather apply to a large dataset?

Option A

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Bob chooses Option B.

Which algorithm would you rather apply to a large dataset?

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Passes through each pair of samples once.



Bob chooses Option B.

Computations on the Million Song Dataset:

10 billion

1 trillion

Why?

Why?

Let n be the number of data points, $f(n)$ be the number of computations.

Why?

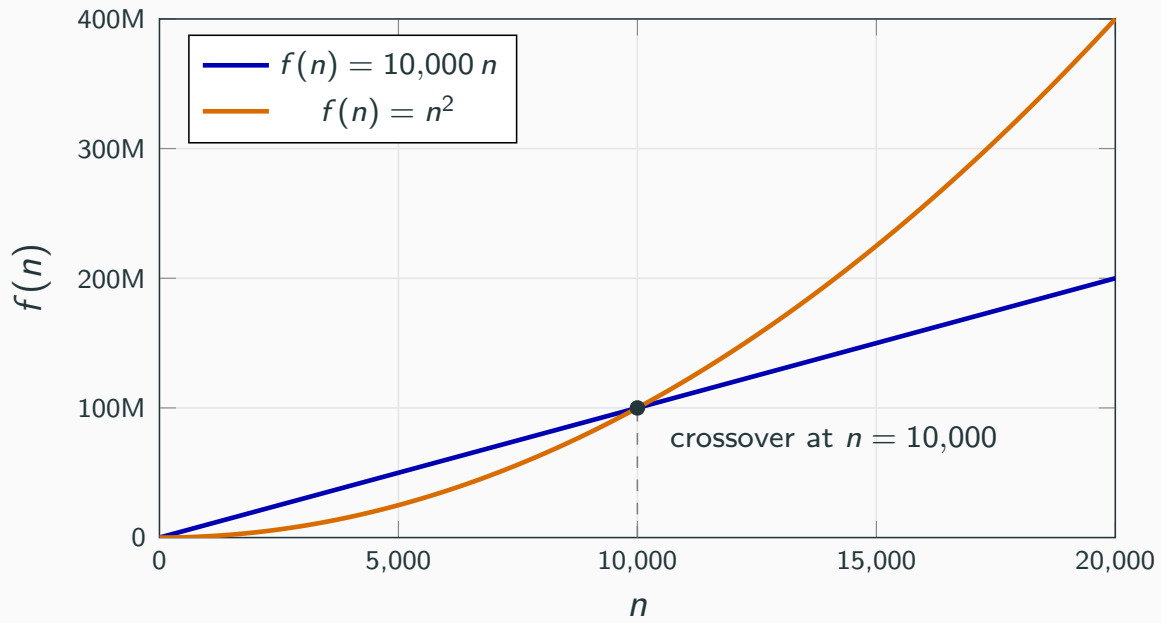
Let n be the number of data points, $f(n)$ be the number of computations.

Option A

$$f(n) = 10,000n$$

Option B

$$g(n) = n^2$$



$g(n) = n^2$ is eventually greater than $f(n) = 10,000n$.

Why?

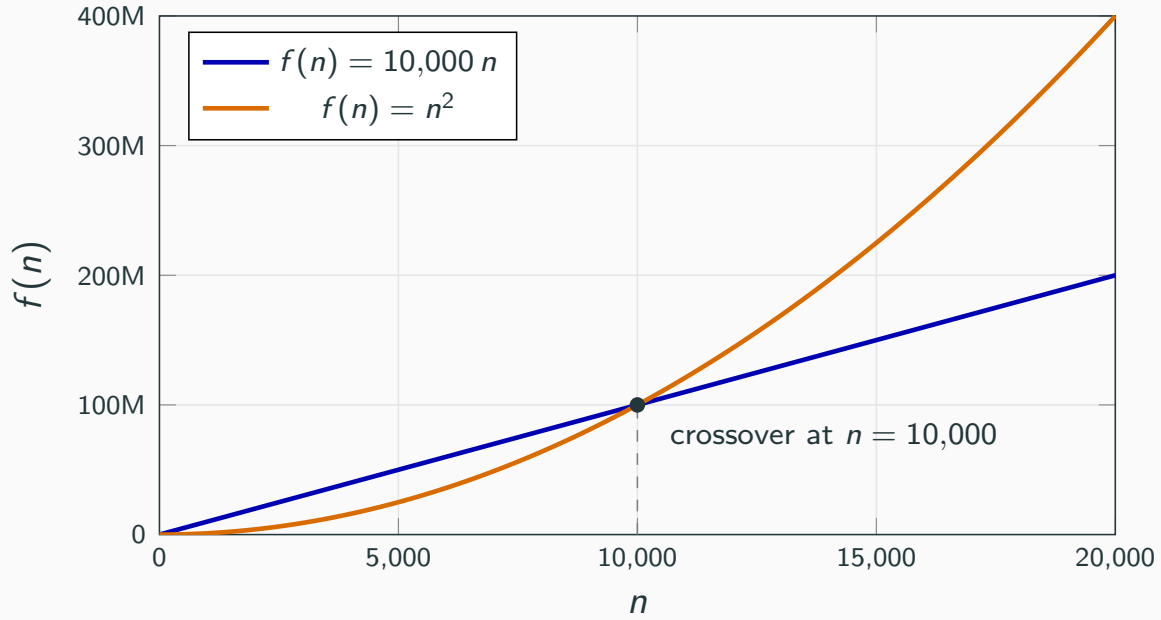
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Option B

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$\underbrace{g(n) = Cn^2}_{\text{quadratic}}$ is eventually greater than $\underbrace{f(n) = Dn}_{\text{linear}}$.

How to compare the efficiency of algorithms?



Figure 1: German mathematician Paul Bachmann

How to compare the efficiency of algorithms?

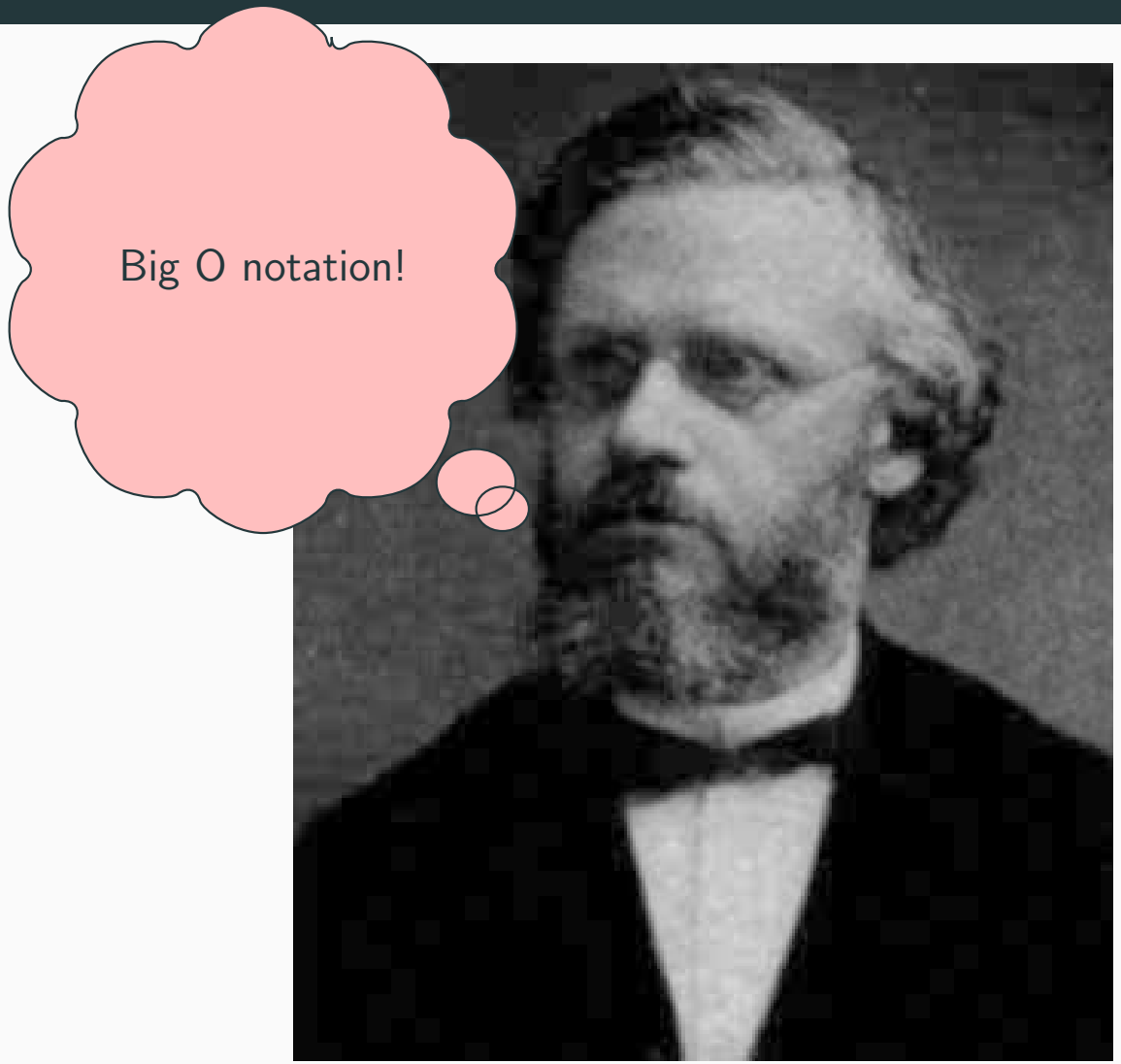


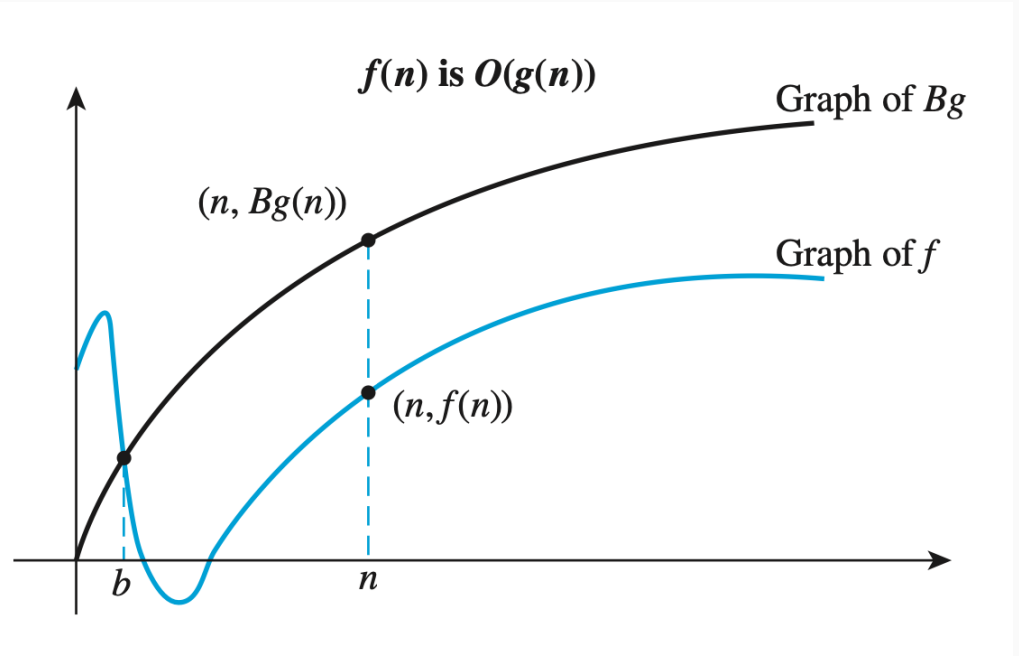
Figure 1: German mathematician Paul Bachmann

Definition

Let f and g be real-valued non-negative functions defined on the same set of nonnegative integers.

Then f is of order at most g, written $\boldsymbol{f(n)}$ **is** $\boldsymbol{O(g(n))}$ (f of n is big-O of g of n) if and only if there exist positive real number B and integer b such that

$$|f(n)| \leq B|g(n)| \quad \text{for every integer } n \geq b$$

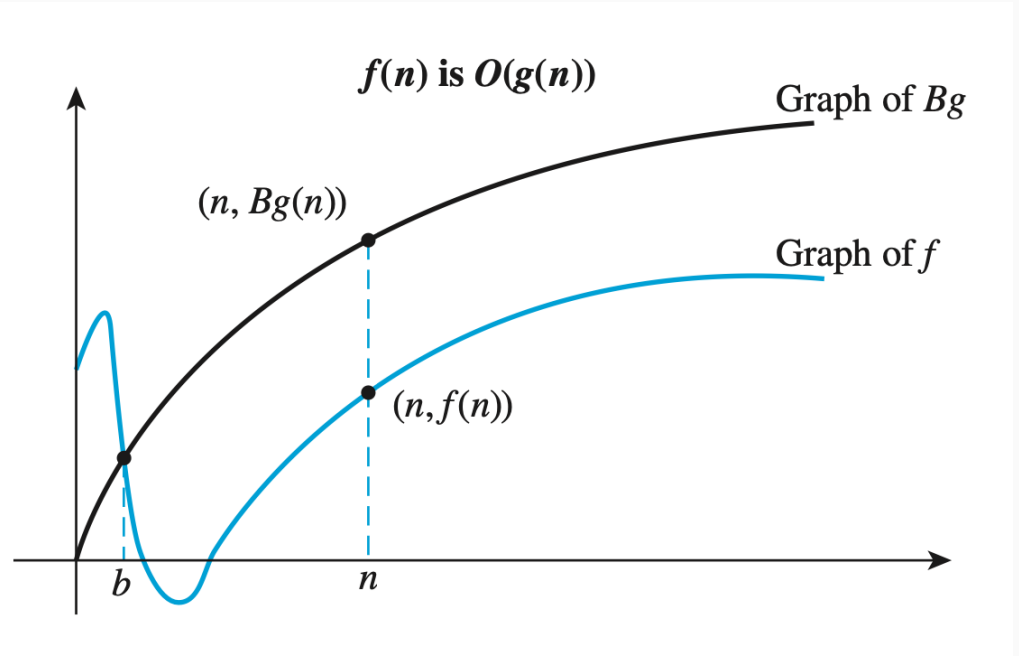


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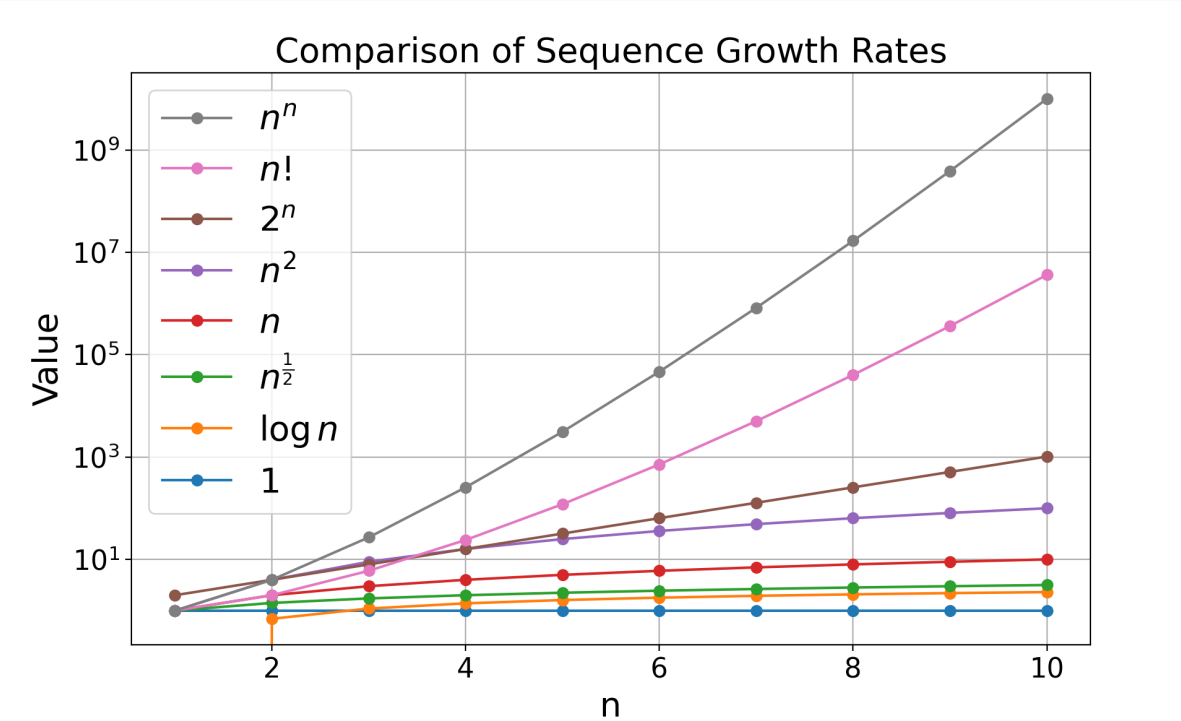
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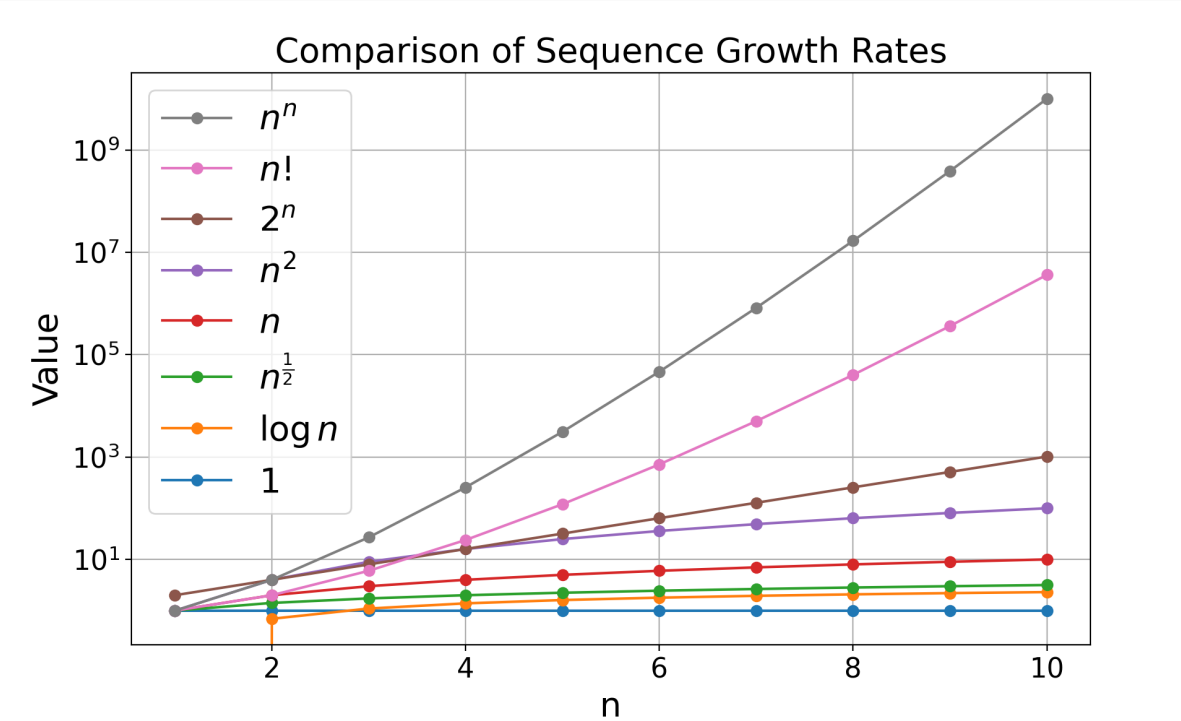
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Classification Theorem



Poll. How should we interpret this graph?

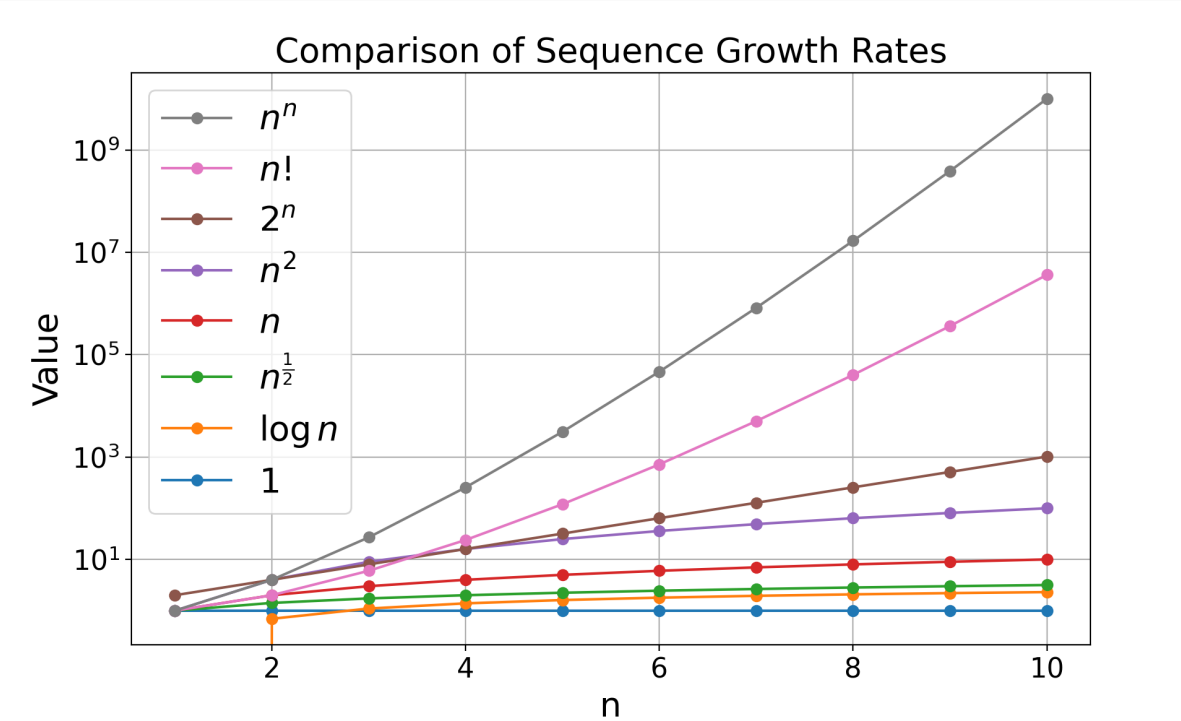
Classification Theorem



Poll. How should we interpret this graph?

- A taxonomy of growth rates (or scaling behavior): Any function is big O of everything equal to or above it, but is **not** big O of anything below it.

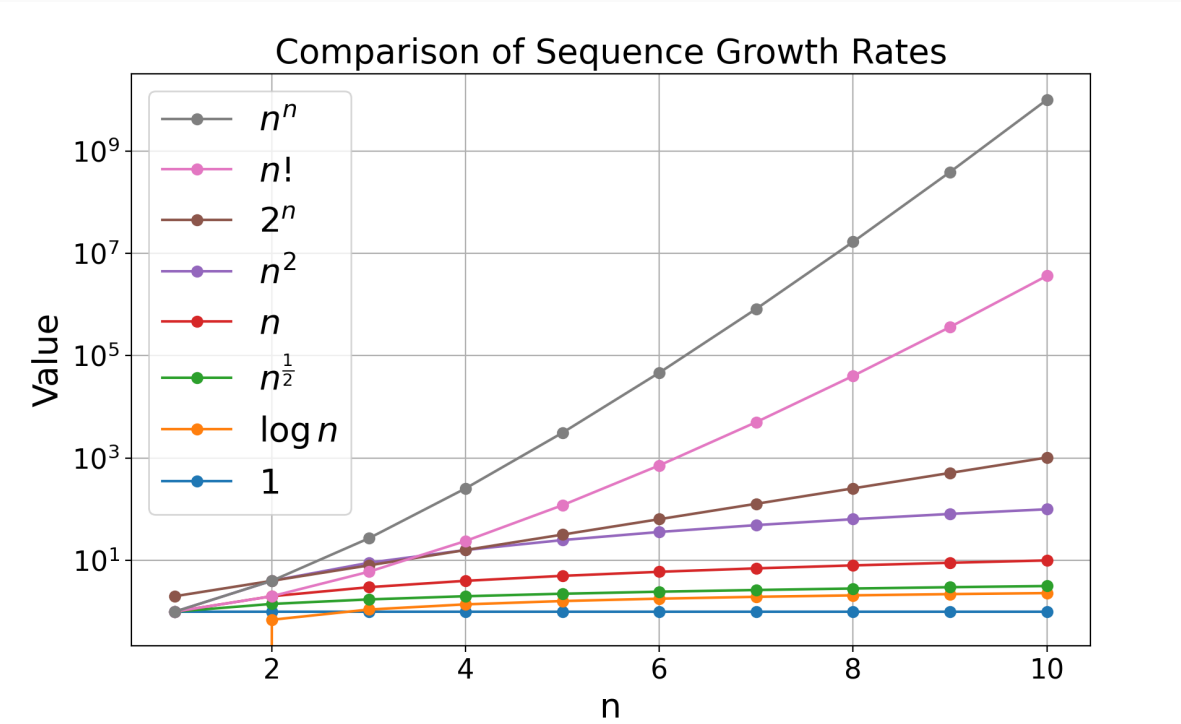
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- **Example**: n is big O of n, n^2, 2^n, $n!$, and n^n. Even if we multiply n by a large constant, those functions will always eventually get bigger!
- **Anti-Example**: n is not big O of $n^{\frac{1}{2}}$. No matter how much you scale up $n^{\frac{1}{2}}$ by a large constant, n will eventually get bigger!

How to check Big O membership?

Example: $f(n) = 3n + 100$ is $O(n)$

Proof. Take $n \geq 1$. Then

$$|3n + 100| \leq |3n + 100n| = 103|n|.$$

(To satisfy the definition of big O, we take $b = 1, B = 103$.)

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$$|9n^2 + 3n - 2| \leq |9n^2 + 3n^2 + 2n^2| = 14|n^2|.$$

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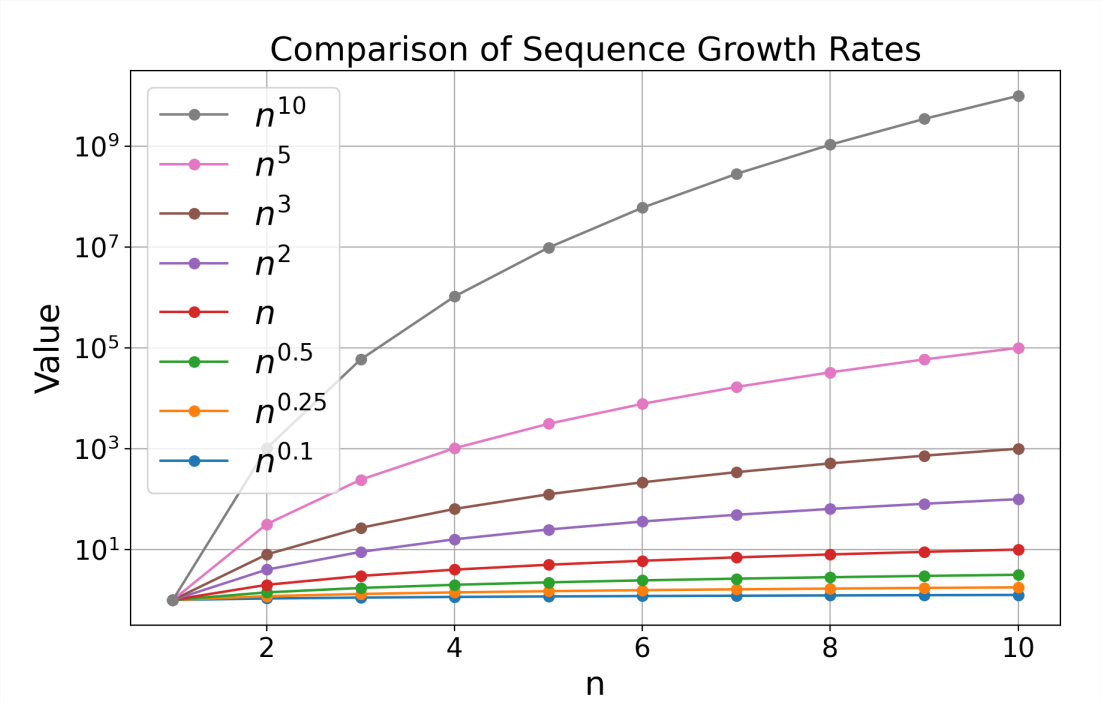
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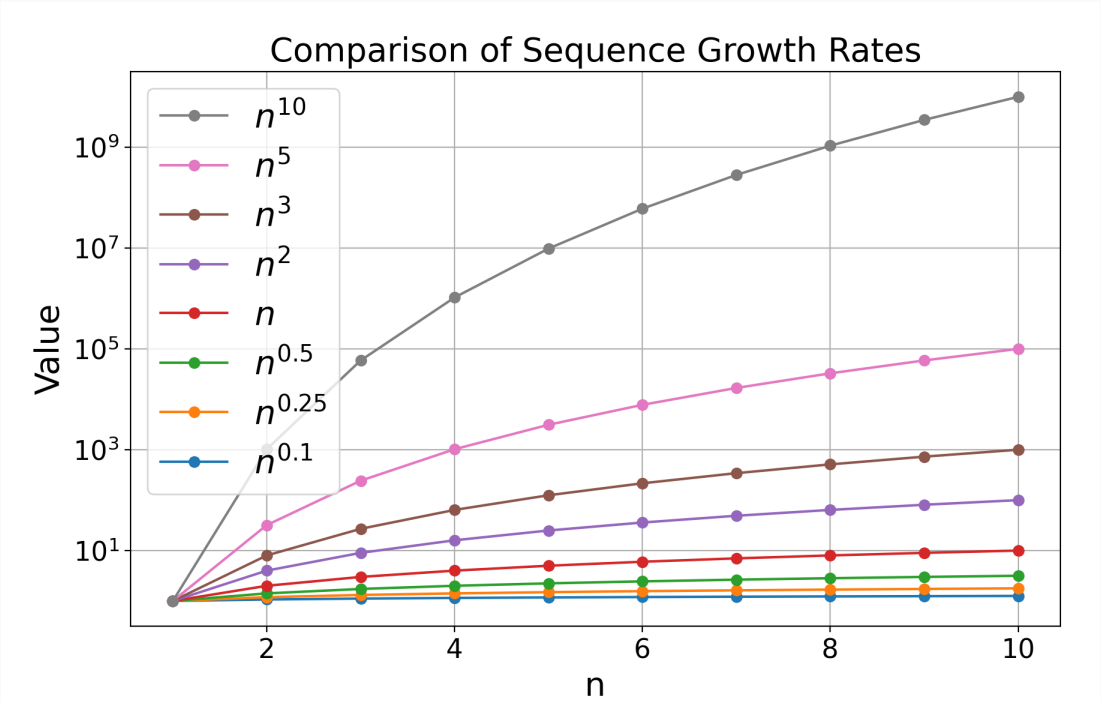
Remark. ...and also big O of still higher-order terms. E.g. for $n \geq 1$,

$$14|n^2| \leq 14|n^3| \leq 14|n^4| \leq \dots$$

Classification Theorem: Zooming In On Polynomials

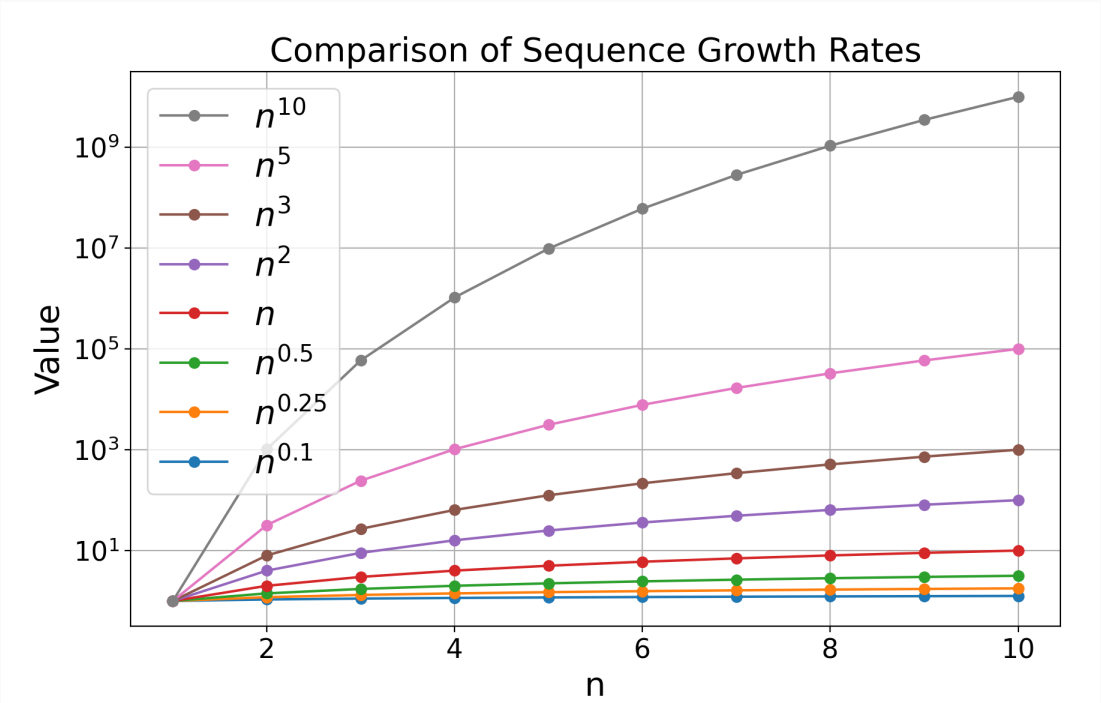


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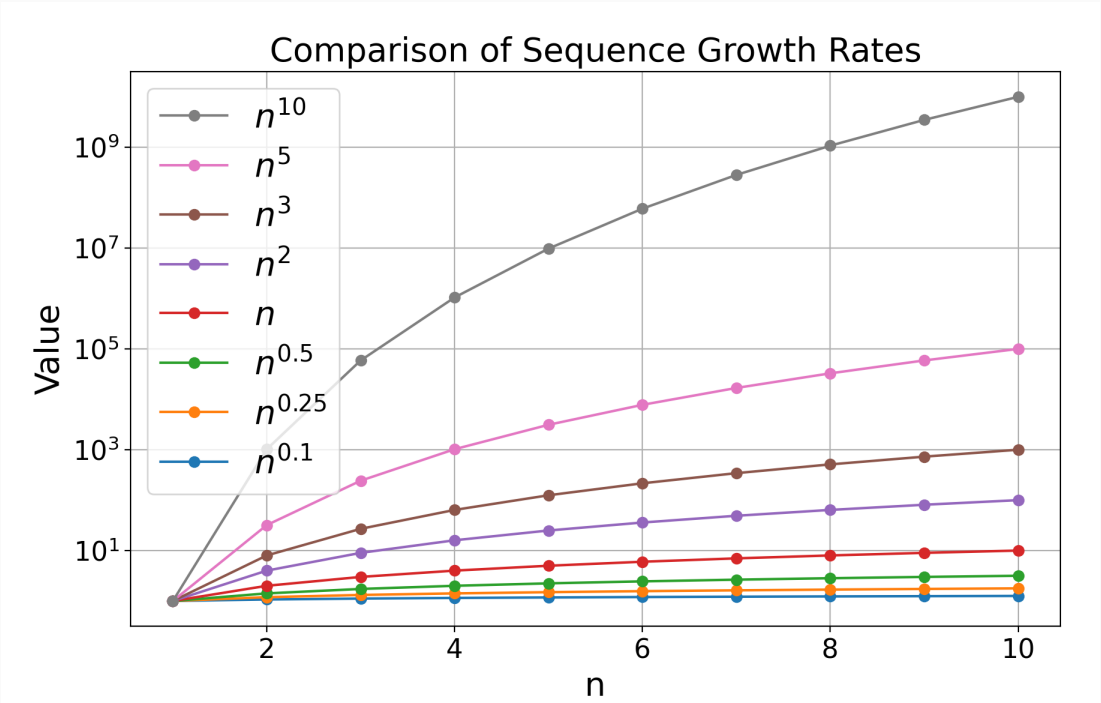
A taxonomy of growth rates: As you go down the legend, each function is big O of everything above it, but **not** big O of everything below it.

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Classification Theorem: Zooming In On Polynomials



A taxonomy of growth rates: As you go down the legend, each function is big O of everything above it, but **not** big O of everything below it.

Theorem

For any function f and positive real numbers u and v with $u < v$,

$$f \text{ is } O(x^u) \implies f \text{ is } O(x^v)$$

A major problem with Big O statements

Poll. Suppose a person is analyzing the efficiency of algorithms, and finds that f is $O(x^5)$ and g is $O(x^4)$. Because $4 < 5$, this person concludes that g has a better algorithmic efficiency than f. Is this person's conclusion correct?

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Solution. No. For instance, take $f(x) = x^2$ and $g(x) = x^3$. Then

f is $O(x^2), O(x^3), O(x^4), O(x^5), \ldots$

and

g is $O(x^3), O(x^4), O(x^5), \ldots$

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Remark. The problem is that the upper bounds for Big O can be needlessly large!

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Remark. The problem is that the upper bounds for Big O can be needlessly large!

We should interpret "g is $O(x^4)$" as: "g is of order **at most** x^4".

How to compare the efficiency of algorithms?



Figure 2: American computer scientist and mathematician Donald Knuth

How to compare the efficiency of algorithms?

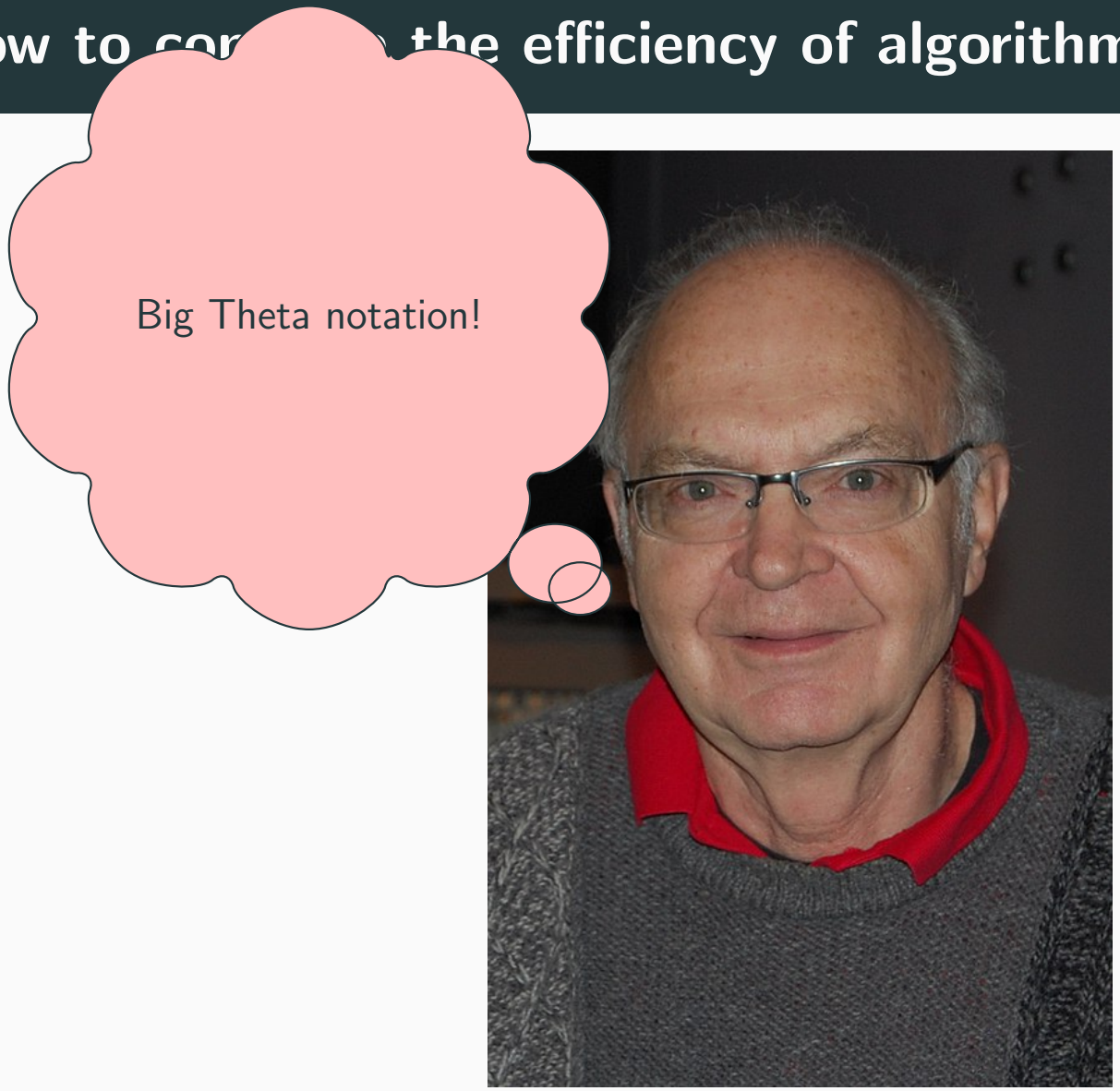
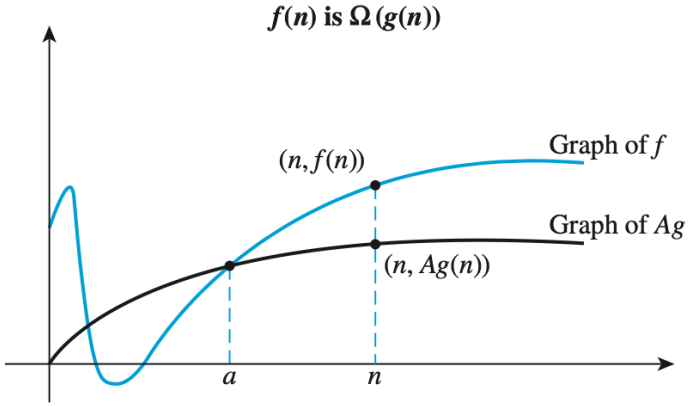
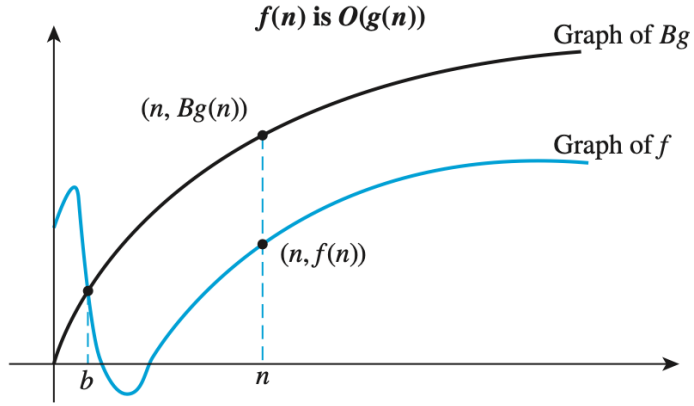


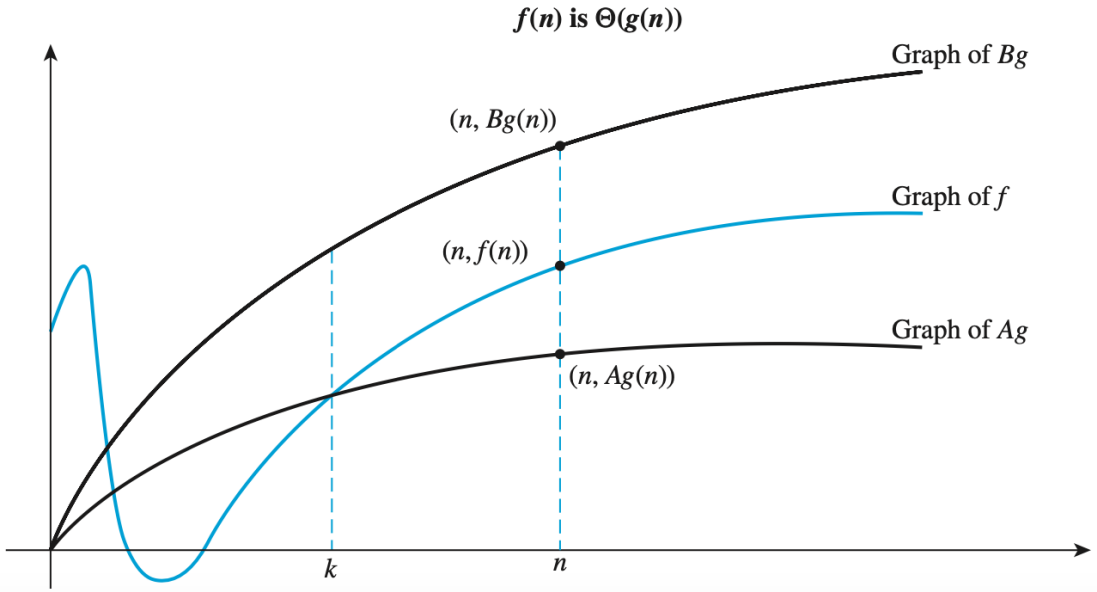
Figure 2: American computer scientist and mathematician Donald Knuth

Overview: Big Theta combines Big O and Big Omega

Big O	f is $O(g)$	f is of order at most g	The values of f are eventually **less** than those of a positive multiple of g.
Big Omega	f is $\Omega(g)$	f is of order at least g	The values of f are eventually **greater** than those of a positive multiple of g.
Big Theta	f is $\Theta(g)$	f is of order g	The values of f are eventually **less** than those of a positive multiple of g and **greater** than those of a positive multiple of g.


$f(n)$ is $\Omega(g(n))$
Graph of f
$(n, f(n))$
Graph of Ag
$(n, Ag(n))$
a
n


$f(n)$ is $O(g(n))$
Graph of Bg
$(n, Bg(n))$
Graph of f
$(n, f(n))$
b
n

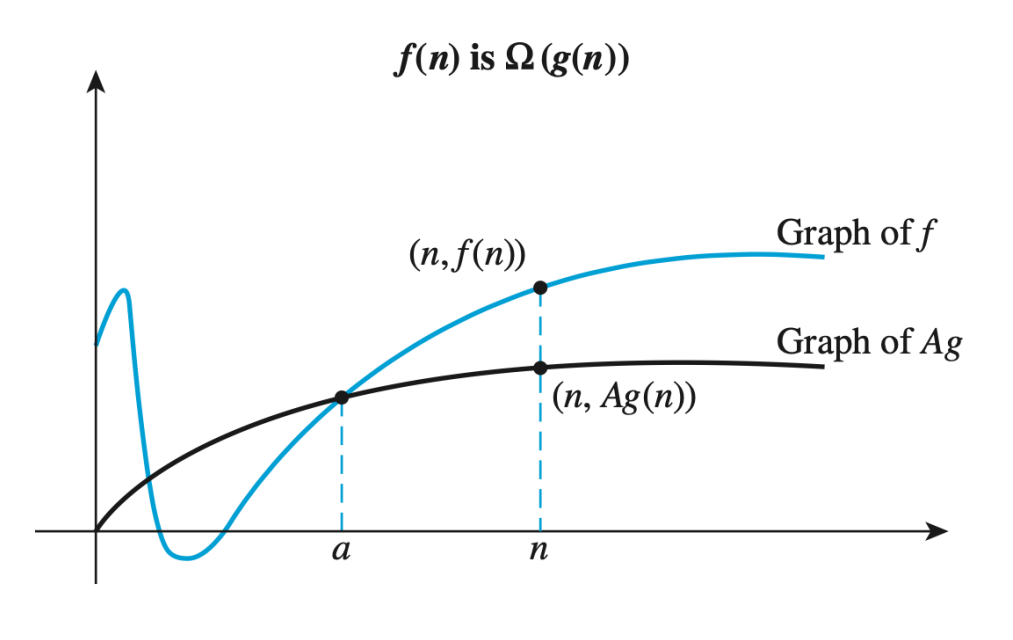

$f(n)$ is $\Theta(g(n))$
Graph of Bg
$(n, Bg(n))$
Graph of f
$(n, f(n))$
Graph of Ag
$(n, Ag(n))$
k
n

Big Omega: Definition

Definition

Let f and g be real-valued non-negative functions defined on the same set of nonnegative integers.
Then f is of order at least g, written $\boldsymbol{f(n)}$ **is** $\boldsymbol{\Omega(g(n))}$ (f of n is big-Omega of g of n) if and only if there exist positive real number A and integer a such that

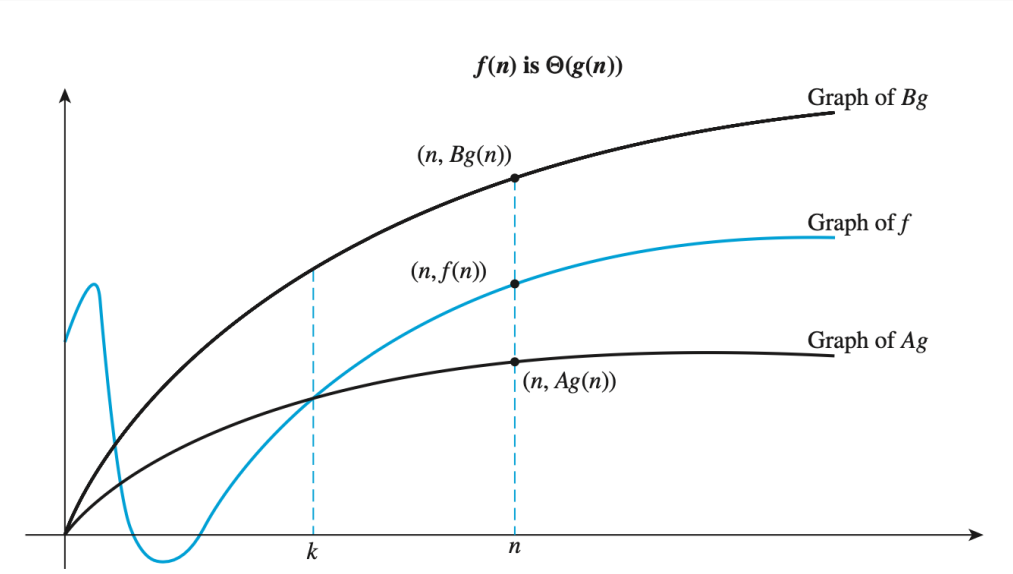
$$|f(n)| \geq A|g(n)| \quad \text{for every integer } n \geq a$$



Big Theta: Definition

Definition

Let f and g be real-valued non-negative functions defined on the same set of nonnegative integers.
Then f is of order g, written $\boldsymbol{f(n)}$ **is** $\boldsymbol{\Theta(g(n))}$ (f of n is big-Theta of g of n) if and only if $\boldsymbol{f(n)}$ is $\boldsymbol{O(g(n))}$ and $\boldsymbol{f(n)}$ is $\boldsymbol{\Omega(g(n))}$.



A useful theorem on Big Theta

Theorem: On Polynomial Orders

If m is any integer with $m \geq 0$ and $c_0, c_1, c_2, \ldots c_m$ are real numbers with $c_m \geq 0$, then

$$c_m x^m + c_{m-1} x^{m-1} + \ldots + c_1 x + c_0 \text{ is } \Theta(x^m).$$

Group exercises

Random group assignments

Ashley, Isabelle: 7
Bailey, Tanna: 3
Blaisdell, Olyver: 7
Brown, Ashton: 10
Cardenas, Dominic: 9
Cutler, George: 12
Fabik, Roland: 6
Fellows, Audrey: 9
Fitzgerald, Ryan: 11
Fuller, Ryan: 3
Griffen, Clara: 1
Grutsch, Tanner: 2
Hager, Nathan: 8
Horner, Jayden: 10
Johnson, Jaret: 5
Kintzel, Samantha: 6
Lameres, Alexis: 2
Lee, Holden: 4
Mayala, Reece: 1
McLean, Connor: 5
Mcbride, Daniel: 1
Miller, Maverick: 9
Moss, Emily: 8
Mousseau, Matthew: 4
Neuman, Evan: 4
Putnam, John: 5
Ramos, Vydel: 12
Rathbun, Jedidiah: 6
Roberts, Matthew: 11
Roller, Nathan: 7
Ryan, Otis: 8
Shevtsov, Timothy: 2
Stensland, Emma: 11
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Group exercises

Exercises.

1. Let $a_n = 1,000,000n + 3,000,000$. Prove that $a_n = O(n)$.
2. Let $a_n = 5 + \frac{1}{n} + \frac{1}{n+1}$. Prove that $a_n = O(1)$.
3. Let $a_n = n^2 + n + 1 + \frac{1}{n} + \sin n$. Prove that $a_n = O(n^2)$.
4. Let $a_n = 3n^2 + 7$. Prove that $a_n = \Theta(n^2)$.
5. Prove Theorem 7.9 from the Ponomarenko textbook.

Note. Please solve all the problems above using the definitions of Big-O and Big-Θ.

Theorem 7.9 (Ponomarenko). Let a_n be a sequence, and b_n, c_n be test sequences. If $a_n = O(b_n)$ and $b_n = O(c_n)$, then $a_n = O(c_n)$.

Solution to Group Exercise #1

Problem. Let $a_n = 1,000,000n + 3,000,000$. Prove that $a_n = O(n)$.

Solution. For $n \geq 1$,

$$|1,000,000n + 3,000,000| \leq |1,000,000n + 3,000,000n| = 4,000,000|n|$$

This satisfies the definition that a_n is $O(n)$ by setting $b = 1$ and $B = 4,000,000$.

Remark. The notation (b and B) comes from the definition of Big O given in this slide deck.

Solution to Group Exercise #2

Problem. Let $a_n = 5 + \frac{1}{n} + \frac{1}{n+1}$. Prove that $a_n = O(1)$.

Solution. For $n \geq 1$,

$$|5 + \frac{1}{n} + \frac{1}{n+1}| \leq |5 + 1 + 1| = 7 \cdot |1|$$

This satisfies the definition that a_n is $O(1)$ by setting $b = 1$ and $B = 7$.

Remark. The notation (b and B) comes from the definition of Big O given in this slide deck.

Solution to Group Exercise #3

Problem. Let $a_n = n^2 + n + 1 + \frac{1}{n} + \sin n$. Prove that $a_n = O(n^2)$.

Solution. For $n \geq 1$,

$$\begin{aligned} |n^2 + n + 1 + \frac{1}{n} + \sin n| &\leq |n^2 + n + 1 + 1 + 1| \\ &\leq |n^2 + n^2 + n^2 + n^2 + n^2| \\ &= 5|n^2| \end{aligned}$$

This satisfies the definition that a_n is $O(n^2)$ by setting $b = 1$ and $B = 5$.

Remark. The notation (b and B) comes from the definition of Big O given in this slide deck.

Solution to Group Exercise #4

Problem. Let $a_n = 3n^2 + 7$. Prove that $a_n = \Theta(n^2)$.

Solution. For $n \geq 1$,

$$|3n^2 + 7| \leq |3n^2 + 7n^2| = 10|n^2|$$

This satisfies the definition that a_n is $O(n^2)$ by setting $b = 1$ and $B = 10$.

Again for $n \geq 1$,

$$|3n^2 + 7| \geq |3n^2| = 3|n^2|$$

This satisfies the definition that a_n is $\Omega(n^2)$ by setting $a = 1$ and $A = 3$.

Since a_n is both $O(n^2)$ and $\Omega(n^2)$, we can conclude that a_n is $\Theta(n^2)$.

Remark. The notation (a, b, A, B) comes from the definitions of Big O and Big Theta given in this slide deck.

Solution to Group Exercise #5

Problem. Prove Theorem 7.9 from the Ponomarenko textbook.

Theorem 7.9 (Ponomarenko). Let a_n be a sequence, and b_n, c_n be test sequences. If $a_n = O(b_n)$ and $b_n = O(c_n)$, then $a_n = O(c_n)$.

Solution. Since $a_n = O(b_n)$, we have that for $n \geq b_1$,

$$|a_n| \leq B_1|b_n| \qquad \text{for some constant } B_1. \tag{1}$$

Since $b_n = O(c_n)$, we have that for $n \geq b_2$,

$$|b_n| \leq B_2|c_n| \qquad \text{for some constant } B_2. \tag{2}$$

Hence for $n \geq \max\{b_1, b_2\}$,

$$|a_n| \overset{\text{Eq. (1)}}{\leq} B_1|b_n| \overset{\text{Eq. (2)}}{\leq} B_1 \cdot B_2|c_n|$$

This satisfies the definition that a_n is $O(c_n)$ by setting $b = \max\{b_1, b_2\}$ and $B = B_1 \cdot B_2$.